MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING WAVE

1/ TYPE DOD-017

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Medium power, low noise, frequency as specified in NAVORD Drawing OD 15440, 4 watts minimum

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec | Ic | Ew | Iw | Eb | Ib | tp | tk | T (base plate) | Barometric pressure, reduced |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | µAdc | Vdc | mAdc | Vdc | mAdc | µs | sec | °C | mmHg |
| Maximum: | 7.6 | -22 | -30 | 3,000 | 5.0 | 2,100 | 45 | 3.3 | --- | 80 | 87 |
| Minimum: | 7.2 | -18 | --- | 2,600 | --- | 1,700 | --- | 1.0 | 60 | --- | --- |

PHYSICAL CHARACTERISTICS:

| | | | |
|---|---|---|---|
| Dimensions: | See figure 1 | Marking: | See note 1 |
| DC and RF connectors: | See figure 1 | Weight: | 2.5 pounds, nominal |
| Cooling: | See note 16 | Mounting position: | Any (secured to a suitable baseplate) |
| Focusing: | Periodic permanent magnet | Magnetic shielding: | See note 13 |
| Shelf life: | See note 9 | Coupling: | See note 14 |

TEST CONDITIONS: See notes 2 and 15

| Parameter: | Ef | If | Ec | Ic | Ew | Iw | Ev | Ib | tk | Pi(rf) | F |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | A | Vdc | mAdc | Vdc | mAdc | Vdc | mAdc | sec | W | GHz |
| Maximum: | 7.5 | 0.44 | --- | --- | --- | 0.65 | 2,000 | 40 | --- | As req'd | F18 |
| Nominal: | 7.4 | 0.43 | -20 | 0 | 2,800 | 0.60 | 1,900 | 35 | 60 | (see | --- |
| Minimum: | 7.3 | 0.42 | --- | --- | --- | 0.55 | 1,800 | 30 | --- | note 3) | F3 (see note 15) |

GENERAL:

Qualification - Required

1/ Replaces: Watkins-Johnson type number W-J 3606 (Part number 290329-001)



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|
| | | | | | MINIMUM | MAXIMUM | |
| | Qualification inspection | | | | | | |
| 1011 | Humidity | 7,10 | No voltages applied | --- | --- | --- | --- |
| --- | Vibration, mechanical | 7 | MIL-STD-202, method 204, test condition A; accel = 2G | --- | --- | --- | --- |
| 1042 | Shock, specified pulse | 7,11 | Test condition A, except 20G; no voltages applied | --- | --- | --- | --- |
| | Quality conformance inspection, part 1 | 4 | | | | | |
| 4289 | Heater current | - | Ef = 7.4 V | If | --- | 2.1 | A |
| 4254 | Saturated rf power output | 2,3,12 | F3 and F18 | Pso | 37 | --- | dBm |
| 4250 | Power output (RF) | 2,3 | Gain compression = 0.25±0.25 dB; F3 and F18 | Po | 34 | --- | dBm |
| 4253 | Gain | 2,3 | Po = 34 dBm; F3 and F18 | G | 36 | --- | dB |
| 4251 | Output power and gain variation (small signal) | 2,3,6 | Swept from F3 to F18 | ΔGss | --- | ±1 | dB |
| 4251 | Output power and gain variation (small signal) | 2,3,6 | Function of temperature | ΔGss | --- | ±1 | dB |
| 4256 | Input and output match | 2,5 | | | | | |
| | Operating band (input and output) | - | F3 to F18 | VSWR | --- | 1.5:1 | --- |
| | Nonoperating band (input and output) | - | F = 4.0 to 7.0 GHz | VSWR | --- | 4:1 | --- |
| 4260 | Noise figure | 2 | F3 and F18 | NF | --- | 22 | dB |
| 4263 | Intermodulation or cross modulation | 2,8 | See figures 3 and 4 | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | |
| 1026 | Temperature range (operating) | 2,7 | TA = 0° to +60°C | --- | --- | --- | --- |
| 1027 | Temperature cycling (nonoperating) | 7 | TA = -40° to +75°C; 10 cycles (min) | --- | --- | --- | --- |
| 1105 | Permanence of marking | - | | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | |
| --- | Life test | - | Group D | t | 1,000 | --- | hrs |
| --- | Life-test end points: | | | | | | |
| 4254 | Saturated rf power output | 2,3, 12 | F3 and F18 | Pso | 37 | --- | dBm |
| 4250 | Power output (RF) | 2,3 | Gain compression = 0.25±0.25 dB; F3 and F18 | Po | 34 | --- | dBm |
| 4253 | Gain | 2,3 | Po = 34 dBm F3 and F18 | G | 36 | --- | dB |
| 4260 | Noise figure | 2 | F3 and F18 | NF | --- | 22 | dB |



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NOTES:

1. A copy of the critical performance characteristics including operating voltages shall be shipped with each tube. The operating voltages shall also be clearly indicated on the tube label and on a small decal at the collector end of each tube. Each tube delivered on the contract shall operate optimally under a single combination of discrete voltages which fall within the ranges shown under absolute ratings. All operating requirements of this specification shall be met under these specific voltages.

2. Apply heater current for 1-minute before simultaneously applying cathode, collector, helix, and anode voltages then set the cathode current to the tube under test (TUT) label value. The tube shall operate within ±3 percent of the operating voltages given on the tube label.

3. Pi(rf) is the power incident to the rf terminals of the TUT and shall be defined as that rf input power required to obtain saturation power output, Pso, at the specified frequency. The TUT shall not be damaged when driven with input powers up to +13 dBm.

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

5. The VSWR and frequency at both input and output shall be measured and recorded at the point of maximum VSWR in the frequency band. The TUT shall operate with an input/output impedence of 50 ohms.

6. The gain variation shall be determined by means of a swept signal technique, or equivalent method. If a swept signal technique is used, the sweep rate shall be such as to assure that the gain at each point is identical to that measured on a single frequency basis. If a single frequency method is used, gain measurements shall be made at frequency increments of no greater than 100 MHz.

7. After performance of the specified test, the TUT shall be capable of satisfying the requirements of all quality conformance inspection, part 1 tests specified herein.

8. With helix supply voltage having noise less than shown on figure 2, the undesired signals generated by amplitude modulation shall be less than the limits described on figure 3, and undesired signals generated by frequency modulation shall be less than the limits described on figure 4.

9. The tubes tested and procured in accordance with this tube specification sheet (TSS), shall be capable of shelf-life for a period of five years after date of shipment from the manufacturer's plant when subjected to the storage conditions specified in MIL-F-18870. Tubes placed in operation during this shelf-life period, shall meet all requirements of this TSS. Tubes found to be defective, within the shelf-life period, shall be replaced by the manufacturer at no cost to the Government, or its contractors.

10. The TUT shall perform as specified herein at relative humidity varying from 5 to 100 percent, including condensation at any temperature up to 50°C (122°F).

11. The TUT shall be operational during the test but shall not be required to meet the electrical parameters. No physical damage shall result from the test and the tube shall be capable of operating within the limits of the specification within 50 ms after removal of the shock impulse.

12. The second harmonic output power shall be at least 10 dB below the fundamental, and all harmonics higher than the second shall be at least 25 dB below the fundamental.

13. The TUT shall be capable of meeting all performance requirements whan a magnetic field, with any orientation and a strength of up to 10 gauss, intersects the tube. The tube shall be capable of meeting all performance requirements with soft magnetic materials as close as 3 inches to any part of the tube.

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NOTES: - Continued

14. The modulation coupling to ac filament current shall be less than 0.1 dB/A.

15. Frequencies F1 thru F18 shall be as specified in NAVORD Drawing OD 15440.

16. Cooling shall be by thermal conduction through an aluminum baseplate. The temperature of the baseplate shall not exceed 80°C.

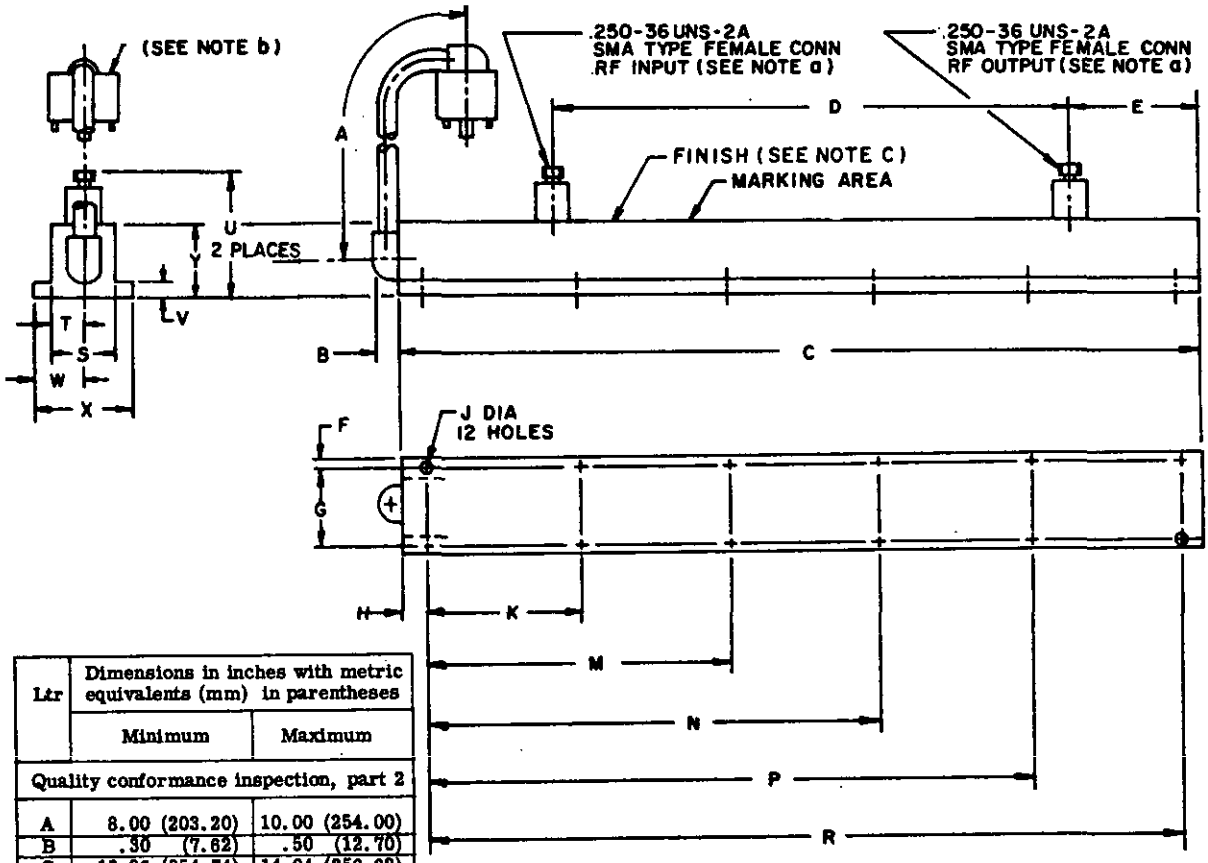
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N062)

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| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| A | 8.00 (203.20) | 10.00 (254.00) |
| B | .30 (7.62) | .50 (12.70) |
| C | 13.96 (354.74) | 14.04 (356.62) |
| D | 8.50 (215.90) | 9.50 (241.30) |
| E | 1.90 (48.26) | 2.60 (66.04) |
| G | 1.370 (34.80) | 1.380 (35.05) |
| J | .184 (4.67) | .191 (4.85) |
| K | 2.620 (66.54) | 2.630 (66.80) |
| M | 5.245(133.22) | 5.255(133.48) |
| N | 7.870(199.90) | 7.880(200.15) |
| P | 10.495(266.57) | 10.505(266.83) |
| R | 13.120(333.25) | 13.130(333.50) |
| S | 1.10 (27.94) | 1.15 (29.21) |
| T | .50 (12.70) | .70 (17.78) |
| U | 2.00 (50.80) | 2.42 (61.47) |
| Y | 1.23 (31.24) | 1.27 (32.26) |
| | Reference dimensions | |
| F | .125 (3.18) | |
| H | .440(11.18) | |
| V | .25 (6.35) | |
| W | .812(20.62) | |
| X | 1.625(41.28) | |

| Pin Allocation | Function | Wire Color |
|---|---|---|
| A | Blank | |
| B | Filament | Brown |
| C | Blank | |
| D | Anode | Green |
| E | Helix/ground | Orange and/or braid |
| F | Thermal | White/orange tracer |
| H | Helix/ground | |
| J | Collector | Red |
| K | Helix/ground | |
| L | Cathode/Filament | Yellow |
| M | Blank | |
| N | Helix | Black |

NOTES:

a. RF connectors. Shall mate with Automatic Metals Products Corporation, P/N E3302-000-819, or equivalent.

b. DC and AC connections. Winchester P/N M12P, 12 pin male connector with M12H hood, or equivalent wired as above.

c. Finish: Cover, blue per FED-STD-595, Color No. 25109; base, chemical film coating per MIL-C-5541 type 1, grade C, class 2.

FIGURE 1. Outline drawing of electron tube type DOD-017.

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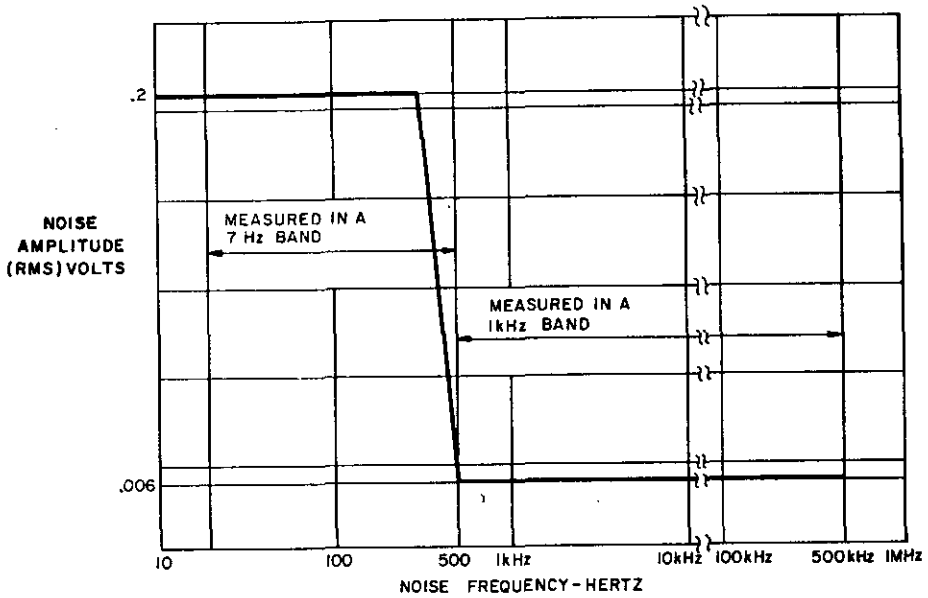



FIGURE 2. Noise frequency - Hertz.

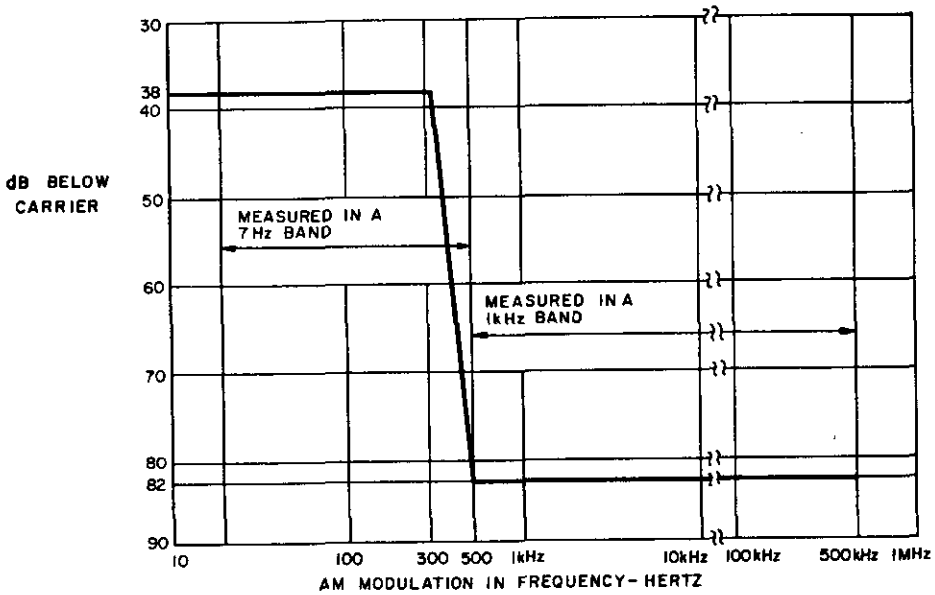



FIGURE 3. AM noise.

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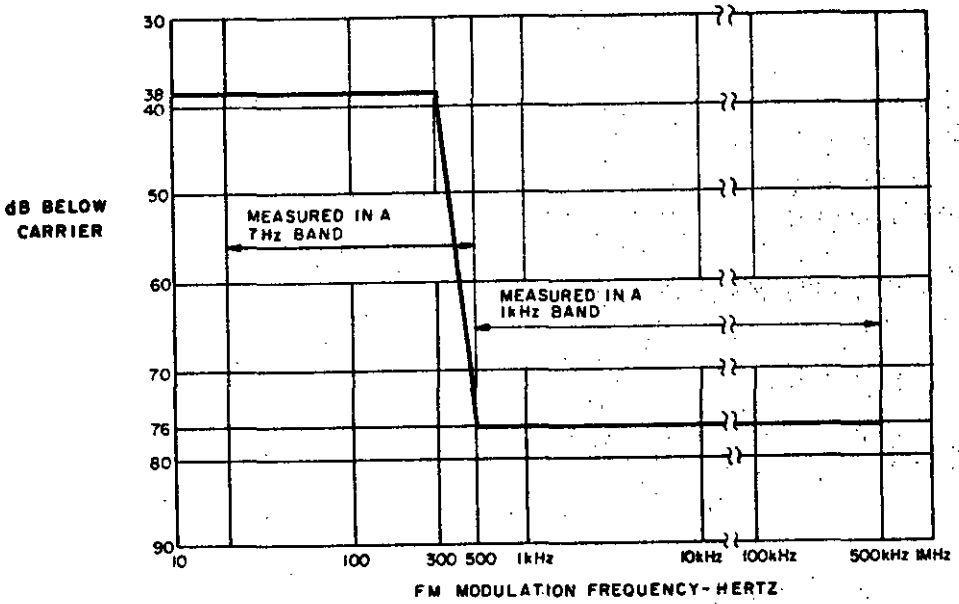



FIGURE 4. FM noise.

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